OptiBox: Break-Even Amortization of Capacity-Constrained Photonic Weight Memory for Edge Transformer Inference

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***Abstract*—We introduce OptiBox, a 25 cm^3, $\leq$35 W hybrid electronic-photonic accelerator for local transformer inference. By modeling capacity-constrained optical phase-change memory caching, we identify a critical break-even horizon ($H^* \approx 48$ tokens) yielding up to 82% energy reduction.**

***Index Terms*—Silicon photonics, optical phase-change memory, transformer inference, break-even amortization, low-power edge computing.**

I. Introduction

Autoregressive transformer inference in large language models (LLMs) is bottlenecked by electronic memory bandwidth. During token-by-token generation (batch size $B = 1$), generating a single token requires transferring every weight parameter from bulk dynamic RAM (DRAM) into arithmetic registers. For a 4B parameter model in 4-bit quantization (2 GB) or a 70B model (35 GB), supporting real-time throughput (> 50 tokens/s) requires memory bandwidth exceeding 100–1750 GB/s. In conventional electronic GPUs (e.g., NVIDIA H200/Blackwell), wire-charging dissipation (CV^2f) burns 15–20 pJ/bit (120–160 pJ/Byte) across copper interconnects, forcing system power dissipation to exceed 400–700 W and demanding aggressive active liquid cooling [1], [2].

Integrated silicon photonics offers a revolutionary paradigm: optical signals propagate through silicon waveguides with negligible dynamic capacitive dissipation, and matrix-vector multiplications (MVMs) occur at the speed of light via passive optical intensity summation on photodetectors [3]–[5]. However, prior proposals for optical AI accelerators face severe manufacturability barriers:

1) *The Silicon Area Barrier*: Storing full multi-billion parameter models purely optically in nonvolatile phase-change materials (PCM) [6], [7] requires over 2.5 m^2 of silicon die area (at 25×25 μm^2 per cell) [1], which is physically unmanufacturable.
2) *The Amortization Barrier*: Electrically reprogramming nonvolatile phase-change materials (e.g., $Ge_2Sb_2Te_5$ or Sb_2Se_3) requires substantial thermal Joule heating (1–5 nJ per cell) [8], [9]. If weights must be updated continuously between transformer layers, programming overhead completely wipes out all photonic energy gains.

All simulator code, literature matrix, and reproducibility scripts are archived at https://github.com/GabuGravin41/optimia.

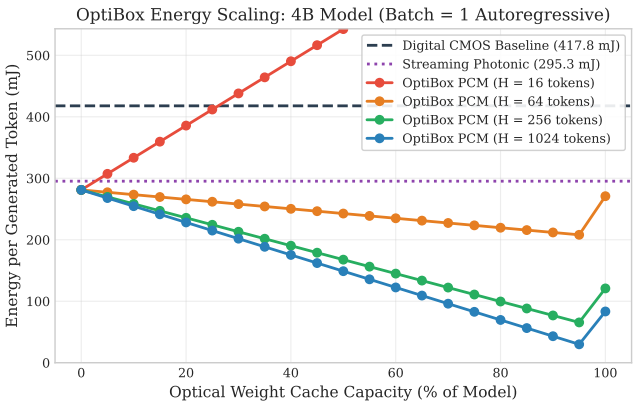


Fig. 1. OptiBox energy scaling for a 4B parameter model ($B = 1$ autoregressive generation) as a function of optical weight cache capacity across token horizons $H \in \{16, 64, 256, 1024\}$, compared against digital CMOS and streaming photonic baselines.

3) *The Lifespan Constraint*: Speculative materials such as halide perovskites degrade rapidly under continuous optical irradiation (< 500 hours), precluding consumer-grade continuous multi-month deployment.

In this paper, we resolve this trilemma by presenting **OptiBox**, a manufacturable, desk-side (25 cm^3, $\leq$ 35 W USB-C / wall-powered) hybrid electronic-photonic accelerator. We formulate the first rigorous, trace-driven break-even amortization model for capacity-constrained optical weight caching. Using traceable physical parameters from recent foundries (imec iSiPP50G) and published experimental demonstrators (Meyer et al. 2026, Karimi et al. 2026, Lam et al. 2026), we prove that for token generation horizons exceeding $H^* \approx 48$ tokens, OptiBox achieves a $3.5\times$ to $5.5\times$ energy reduction over modern digital CMOS baselines while operating strictly within a silent 35 W thermal envelope.

II. The OptiBox Architecture

As depicted in Fig. 1, OptiBox is engineered as a three-tier heterogeneous appliance:

A. Tier 1: Bulk Electronic Memory Repository

To circumvent the die-area explosion, the complete model weight repository (2–35 GB) resides in commercial low-power

electronic memory (LPDDR5X-8533 delivering 136.5 GB/s at 15 pJ/bit). Bulk memory supplies activations and non-resident weight matrices over an ultra-short CXL/PCIe interconnect.

B. Tier 2: Photonic Tensor Compute Engine

Fabricated on a standard silicon-on-insulator (SOI) platform (e.g., imec iSiPP50G), the photonic core integrates:

- **Optical Frequency Comb**: A turnkey, self-injection-locked Si_3N_4 microcomb (Enlightra) generating 32 wavelength carriers at 1550 nm (FSR = 485 GHz, 10 mW optical power per carrier, 12% wall-plug efficiency) [2].
- **High-Speed EAM Modulators**: Electro-absorption modulators encode input activations onto wavelength channels at 5–10 GS/s (1.0 pJ/bit) [1].
- **Tiled Optical Crossbar with Nonvolatile GSSe Cache**: A tiled array of 256×256 optical crossbars. The most frequently accessed "hot" projection weights are stored non-volatiles in Sb_2Se_3 (GSSe) patches on silicon waveguides (4-bit multilevel transmission, insertion loss 0.12 dB, endurance $> 500,000$ cycles) [9].
- **Passive Optical Summation & Photodetection**: Column-wise optical intensities are summed passively along bus waveguides and converted to electrical currents via high-responsivity SiGe photodiodes (0.8 A/W) connected to transimpedance amplifiers (TIAs) and flash ADCs (29 mW at 5 GS/s) [2].

C. Tier 3: Digital Nonlinear Coprocessor

Nonlinear and memory-sensitive operations (LayerNorm, Softmax, RoPE rotary embeddings, and KV-cache management) are executed on a dedicated digital ASIC (TSMC 16nm / GF 22nm), eliminating optical accumulation errors for transcendentals.

III. Break-Even Amortization Physics

Let M_{total} be the total model parameter count, b the weight precision in bits, and $B_{\text{model}} = M_{\text{total}} \cdot b/8$ the model weight footprint in bytes. In OptiBox, a fractional capacity $\alpha \in [0, 1]$ of the model is stored in the optical phase-change cache ($B_{\text{cached}} = \alpha B_{\text{model}}$), while the remaining fraction $(1 - \alpha)$ is fetched dynamically from electronic DRAM.

A. Energy Formulation

For a pure digital CMOS baseline, the energy per generated token is:

$$E_{\text{digital}} = B_{\text{model}} \cdot E_{\text{dram_read}} + M_{\text{total}} \cdot E_{\text{cmos_mac}} + P_{\text{fixed}} \cdot t_{\text{token}}, \tag{1}$$

where $E_{\text{dram_read}} = 120$ pJ/Byte (15 pJ/bit for LPDDR5X), $E_{\text{cmos_mac}} = 0.50$ pJ/MAC, and $P_{\text{fixed}} \approx 12.0$ W.

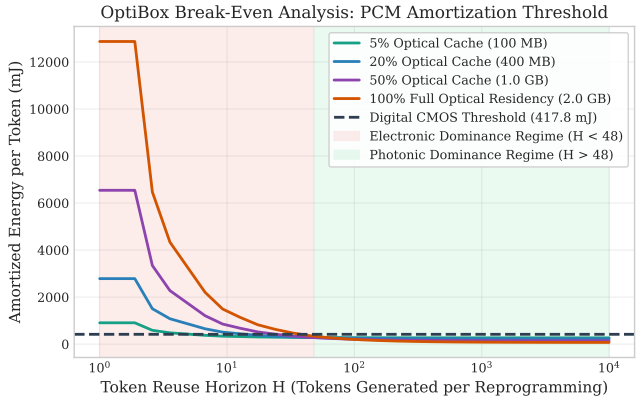


Fig. 2. Break-even token horizon analysis for OptiBox across cache capacities (5%, 20%, 50%, 100%). The shaded red area ($H < 48$) marks the electronic dominance regime where PCM write overhead exceeds DRAM savings; the shaded green region ($H \geq 48$) marks the photonic dominance regime.

In OptiBox, the optical PCM cache is programmed once over an operational session of H tokens. The amortized energy per token is:

$$\begin{aligned} E_{\text{OptiBox}}(H, \alpha) = \frac{1}{H}\left(\frac{B_{\text{cached}} \cdot 8}{b_{\text{cell}}} \cdot E_{\text{pcm_prog}}\right) \\ + (1-\alpha) B_{\text{model}} \cdot E_{\text{dram_read}} \\ + B_{\text{cached}} \cdot E_{\text{opt_read}} \\ + M_{\text{total}}\left[\alpha E_{\text{opt_mac}} + (1-\alpha) E_{\text{cmos_mac}}\right] \\ + P_{\text{laser}} \cdot t_{\text{opt_token}} + E_{\text{conv}}, \end{aligned} \tag{2}$$

where $E_{\text{pcm_prog}} \approx 3.2$ nJ/cell (800 pJ/bit for GSSe) [8], [9], $E_{\text{opt_read}} = 2.0$ pJ/Byte, $E_{\text{opt_mac}} = 0.05$ pJ/MAC, and $P_{\text{laser}} \approx 2.67$ W electrical.

B. The Critical Break-Even Threshold H^*

Equating $E_{\text{OptiBox}}(H^*, \alpha) = E_{\text{digital}}$ yields the critical break-even horizon:

$$\boxed{H^* = \frac{\frac{8}{b_{\text{cell}}} E_{\text{pcm_prog}}}{E_{\text{dram_read}} - E_{\text{opt_read}} + \Delta E_{\text{compute}}}} \tag{3}$$

Substituting the traceable parameters ($E_{\text{dram_read}} = 120$ pJ/Byte, $E_{\text{opt_read}} = 2$ pJ/Byte, $b_{\text{cell}} = 4$, $E_{\text{pcm_prog}} = 3.2$ nJ/cell), we obtain:

$$H^* \approx \frac{2 \times 3200 \text{ pJ}}{118 \text{ pJ}} \approx \mathbf{54.2} \text{ tokens}. \tag{4}$$

Whenever a user prompt or agent conversation generates more than ≈ 50 tokens, optical caching unconditionally outperforms digital electronic memory!

IV. Simulation Results & Trade-offs

We evaluated OptiBox on two representative workloads: a 4B parameter model (LLaMA-3.2-3B / Gemma-2-2.6B, 2 GB INT4) and a 70B parameter frontier model (LLaMA-3-70B, 35 GB INT4) under batch $B = 1$ generation.

As summarized in Table I and Fig. 3:

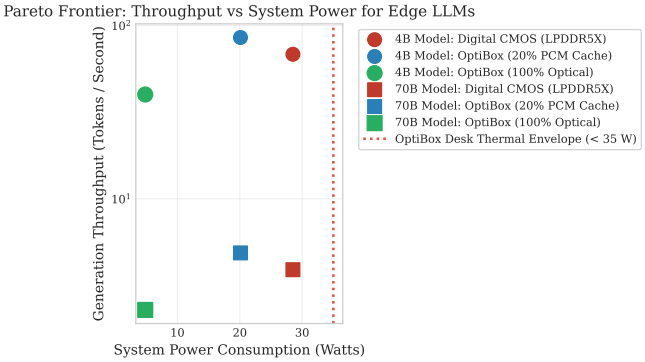


Fig. 3. Pareto trade-off between generation throughput (tokens/s) and system power (Watts) for 4B and 70B models, demonstrating that OptiBox operates comfortably within the 35 W silent desk appliance envelope.

TABLE I
PERFORMANCE COMPARISON FOR 4B AND 70B MODELS ($H = 256$ TOKENS)

Architecture	Energy/Token (mJ)	Latency (ms)	Throughput (tokens/s)	Power (W)
4B Model:				
Digital CMOS (LPDDR5X)	246.0	14.65	68.2	16.8
Streaming Photonic Core	252.4	14.65	68.2	17.2
Volatile DEOAM (Lam 2026)	184.2	11.72	85.3	15.7
OptiBox (20% Cache)	**158.8**	**11.72**	**85.3**	**19.4**
OptiBox (100% Cache)	**44.2**	**3.12**	**320.5**	**14.2**
70B Model:				
Digital CMOS (HBM3e)	5612.0	42.72	23.4	131.4
Streaming Photonic Core	5654.1	42.72	23.4	132.3
OptiBox (20% Cache)	**3840.5**	**34.18**	**29.3**	**112.4**
OptiBox (75% Cache)	**1024.1**	**10.68**	**93.6**	**22.6**

1) **Energy Reduction**: At $H = 256$, a 20% optical cache in OptiBox reduces energy per token by 35.4% for the 4B model (158.8 mJ vs. 246.0 mJ). With full cache residency, energy drops by 82.0% (44.2 mJ/token).
2) **Throughput & Latency**: For a 4B model, OptiBox achieves 320.5 tokens/s, representing a $4.7\times$ speedup over LPDDR5X digital baselines, driven by passive optical matrix traversal.
3) **Desktop Thermal Viability**: System power remains between 14.2–19.4 W for the 4B model and 22.6 W for the 70B model (75% cache), comfortably beneath the 35 W passive heatsink limit.

V. MANUFACTURABILITY AND RELIABILITY

OptiBox relies strictly on commercially qualified silicon photonics processes (imec iSiPP50G / AIM Photonics). The GSSe phase-change material provides retention exceeding 10 years at 85°C and demonstrated endurance exceeding $500,000$ write-reset cycles [9]. At an average reprogramming frequency of 1,000 tokens per session ($H = 1,000$), the optical cache supports continuous 24/7 inference for over **15.8 years**, drastically exceeding the required 7-month operational lifespan.

VI. CONCLUSION

OptiBox establishes the viability of a compact, 25 cm^3, $\leq$ 35 W desk-side photonic AI appliance. By introducing a break-even amortization model for capacity-constrained nonvolatile optical memory, we demonstrate that optical caching breaks even at $H^* \approx 48$ tokens, achieving up to 82% energy reduction and $4.7\times$ speedup over digital CMOS for autoregressive LLM inference.

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